MATH607 - Algebraic Geometry II

Notes taken by Haoran Li

Department of Mathematics
University of Maryland

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1 HW 1 Solutions

Problem 1.1 First we have a lemma: If A is a Noetherian commutative ring, $\mathfrak{p}$ is a prime ideal, M is a finitely generated module, and $M_{\mathfrak{p}}$ is a free module over $A_{\mathfrak{p}}$, then there exists $f \in A - \mathfrak{p}$ such that M_f is a free module over A_f

1. Suppose $\mathfrak{p} = x \in V = \text{Spec } A$, where A is Noetherian, $\mathcal{F}|_V = \widetilde{M}$, then $\mathcal{F}_x = \widetilde{M}_{\mathfrak{p}}$, $\mathcal{O}_{X,x} = A_{\mathfrak{p}}$, apply lemma, we know that $x \in U = \text{Spec } A_f$ for some $f \in A - \mathfrak{p}$ such that $\mathcal{F}|_U \cong \widetilde{M}_f$ is free
2.
3. Consider sheaf $\mathcal{G} = \text{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{O}_X)$. Define $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G} \to \mathcal{O}_X$, locally, it reads $M \otimes_A \text{Hom}(M, A) \to A$, $m \otimes_A f \mapsto f(m)$. M is a rank one free A module, so this map is an isomorphism locally, hence an isomorphism of sheaves

Problem 1.2

1.
2. If $\mathcal{F}$ is locally free, then ϕ is locally constant
3.

Problem 1.3

1. The normalization is given by $\phi : \text{Spec } k[t] = \tilde{X} \to X$, which is defined through $f : k[x, y]/(y^2 - x^3) \to k[t]$, $x \mapsto t^2, y \mapsto t^3$
2.

Problem 1.4

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2.
3.
4.
5.

Problem 1.5

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2.

Problem 1.6

Problem 1.7

1.
2.
3.
4.
5.

Problem 1.8 Consider $k' = k[t^{\frac{1}{p}}]$, , then over k', X is defined by $y^2 = x^p - t = (x - t^{\frac{1}{p}})^p$

Problem 1.9

2 HW 2 Solutions

Problem 2.1

1.

Problem 2.2

1.

Problem 2.3

1.

Problem 2.4

1.

Problem 2.5

1.

Problem 2.6

1.

Problem 2.7

1.

3 HW 3 Solutions

Problem 3.1

1.

Problem 3.2

1.

Problem 3.3

1.

Problem 3.4

1.

Problem 3.5

1.

Problem 3.6

1.

Problem 3.7

1.

4 HW 4 Solutions

5 HW 5 Solutions

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